

HYPERTENSION DIET GUIDELINES

Low-Sodium, Heart-Healthy DASH Diet Clinical Reference

1. What is Hypertension?

Hypertension (high blood pressure) is defined as sustained systolic BP ≥ 130 mmHg and/or diastolic BP ≥ 80 mmHg (ACC/AHA 2017) or $\geq 140/90$ mmHg (WHO/JNC-7). It affects ~1.28 billion adults worldwide and is the leading modifiable risk factor for stroke, myocardial infarction, heart failure, and chronic kidney disease.

Classification	Systolic (mmHg)	Diastolic (mmHg)	Action
Normal	<120	<80	Maintain healthy lifestyle
Elevated	120–129	<80	Lifestyle modification
Stage 1 HTN	130–139	80–89	Lifestyle + consider medication
Stage 2 HTN	≥ 140	≥ 90	Medication + intensive lifestyle changes
Hypertensive Crisis	>180	>120	Immediate medical attention

2. DASH Diet Overview (CRITICAL)

The DASH (Dietary Approaches to Stop Hypertension) diet is the most evidence-based dietary pattern for BP reduction. Clinical trials show DASH reduces systolic BP by 8–14 mmHg — comparable to a single antihypertensive medication. The diet is endorsed by AHA, NHLBI, WHO, and ICMR as a first-line lifestyle intervention.

Food Group	Recommended Servings/Day	DASH Goal	Example Serving
Vegetables	4–5 servings/day	Potassium, magnesium, fiber	1 cup raw or 1/2 cup cooked
Fruits	4–5 servings/day	Potassium, magnesium, fiber	1 medium fruit or 1/2 cup
Whole grains	6–8 servings/day	Fiber, energy	1 slice bread or 1/2 cup cooked
Low-fat dairy	2–3 servings/day	Calcium, protein	1 cup milk or yogurt
Lean meat/fish/poultry	≤ 6 oz/day	Protein, magnesium	3 oz cooked = palm-size
Nuts, seeds, legumes	4–5 servings/week	Potassium, protein, Mg	1/3 cup nuts or 1/2 cup beans
Fats and oils	2–3 servings/day	Healthy fats	1 tsp oil or 1 tbsp mayo
Sweets/added sugars	≤ 5 servings/week	Limit	1 tbsp sugar or 1 small candy

3. Sodium Guidelines

Target	Sodium Limit	Estimated BP Reduction	Equivalent Salt
Standard DASH	<2300 mg/day	2–4 mmHg systolic	~1 tsp table salt/day
Low-Sodium DASH	<1500 mg/day	4–6 mmHg systolic	~3/4 tsp table salt/day
Indian avg. intake (est.)	~3500–4500 mg/day	—	~2–2.5 tsp/day (exceeds target)

High-sodium foods to limit:

- Pickles (achar): 1 piece = 200–500 mg sodium
- Papad: 1 medium = 150–250 mg
- Packaged namkeen, chips: 300–500 mg/serving
- Processed/canned foods (soups, sauces): 400–800 mg/serving
- Instant noodles (1 packet): ~1000–1500 mg
- Salted butter, cheese, processed meats
- Restaurant meals (typically 1000–2000 mg per meal)

4. Potassium — Critical Counter to Sodium

Potassium helps kidneys excrete excess sodium and relaxes blood vessel walls. **Target: 3500–4700 mg/day** from food (NOT supplements unless prescribed; dangerous in CKD/hyperkalemia). Each 1000 mg/day increase in potassium reduces systolic BP by ~1.5–3 mmHg.

Food	Potassium (mg)	Serving	Notes
Banana	~422 mg	1 medium	Easy daily source
Spinach (cooked)	~839 mg	1 cup	Best vegetable source
Sweet potato (baked)	~694 mg	1 medium	Also fiber, beta-carotene
Avocado	~975 mg	1 fruit	MUFA + potassium
Coconut water	~600 mg	1 cup (240ml)	Natural electrolyte
Rajma (cooked)	~713 mg	1 cup	Protein + potassium
Salmon/tuna	~534 mg	100g	Omega-3 bonus
Yogurt (plain)	~380 mg	1 cup	Daily inclusion recommended
Pomegranate	~666 mg	1 cup seeds	Antioxidant + potassium
Tomatoes	~292 mg	1 cup	Also lycopene for CVD

5. Food Categories: Recommended vs. Avoid

■ RECOMMENDED

- All fresh vegetables, especially leafy greens
- Fresh fruits (banana, pomegranate, citrus)
- Whole grains: oats, brown rice, barley, millets
- Low-fat dairy: yogurt, buttermilk, skimmed milk
- Legumes: lentils, rajma, chickpeas, moong
- Lean protein: fish (2–3×/wk), grilled chicken, egg whites
- Nuts (unsalted): almonds, walnuts (small portions)
- Herbs/spices: garlic, ginger (natural BP lowerers)
- Olive/canola oil (replace butter/vanaspati)
- Potassium-rich coconut water (1 cup/day)

■ AVOID / RESTRICT

- Table salt, rock salt, sea salt in excess
- Pickles, papads, chutneys (high sodium)
- Packaged snacks, chips, namkeen, biscuits
- Processed meats: sausages, salami, bacon
- Canned/preserved foods (soups, sauces, ketchup)
- Instant noodles, fast food, restaurant meals (frequently)
- Full-fat cheese, butter, cream, vanaspati
- Alcohol >1–2 units/day (raises BP)
- Caffeinated beverages in excess (>3 cups/day)
- High-fat red meat, organ meats

6. Fat Guidelines

- **Total fat:** 25–35% of total daily energy.
- **Saturated fat:** <7% of total energy. Limit butter, ghee, coconut oil, red meat.
- **Trans fat:** <1% or ideally ZERO. Avoid vanaspati, partially hydrogenated oils, bakery fat.
- **MUFA/PUFA:** Favor olive oil, mustard oil (1–2 tsp/meal), walnuts, fatty fish.
- **Omega-3 fatty acids** (fish, flaxseed, walnuts) reduce triglycerides and vascular inflammation.
- Use **cooking spray or minimal oil** for Indian cooking; steaming, baking, air-frying preferred over deep-frying.

7. Sample DASH Diet Meal Plan — Indian Context (2000 kcal)

Meal	Time	Items	Na (approx)
Breakfast	8:00 AM	Oats (50g) with banana + skimmed milk (200ml) + 5 almonds (unsalted)	<100mg
Mid-Morning	10:30 AM	Coconut water (200ml) + 1 guava	<30mg
Lunch	1:00 PM	Brown rice (1 cup) + dal (no added salt, use lemon) + spinach sabzi (minimal salt) + curd + salad	~400mg
Evening	5:00 PM	Roasted makhana (unsalted, 30g) + herbal tea	<50mg
Dinner	7:30 PM	2 jowar roti + rajma (low-salt) + tomato-cucumber raita	~350mg
Total Sodium		~930 mg (well within 1500 mg target)	■

8. Lifestyle Advice

Weight Control: Losing 1 kg body weight reduces systolic BP by ~1 mmHg. Target BMI 18.5–24.9.

Physical Activity: 30 min moderate aerobic exercise 5×/week (brisk walking, cycling) reduces BP by 4–9 mmHg.

Alcohol Reduction: Limit to ≤ 1 drink/day (women) and ≤ 2 drinks/day (men). Excess alcohol raises BP significantly.

Smoking Cessation: Each cigarette raises BP transiently; chronic smoking hardens arteries. Cessation is critical.

Stress Management: Chronic stress activates RAAS (renin-angiotensin system), raising BP. Yoga, meditation, and breathing exercises have evidence.

Sleep: 6–8 hours/night. Sleep apnea and short sleep duration are independent HTN risk factors.

References: NHLBI DASH Eating Plan • ACC/AHA Hypertension Guidelines 2023 • WHO Global Hypertension Guidelines 2021 • ICMR Dietary Guidelines 2020